

HAMILTON INTERNATIONAL HOTEL

DIRE DAWA, ETHIOPIA



ENTERPRISE HOTEL MANAGEMENT SYSTEM

DIGITAL TRANSFORMATION & IMPLEMENTATION PROPOSAL

Proposal Vision

A secure, integrated and scalable platform connecting reservations, front desk, rooms, housekeeping, finance, food and beverage, guest relationships, revenue management, staff operations and management reporting.

Prepared for	Hamilton International Hotel
Location	Dire Dawa, Ethiopia
Proposal version	Version 1.0
Date	July 2026

SMART OPERATIONS | EXCEPTIONAL GUEST EXPERIENCE | CONTROLLED GROWTH

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PROPOSAL NOTICE

Confidentiality, purpose and document control

Confidentiality Statement

This document contains a proposed digital operating model, system architecture, implementation approach and commercial planning assumptions for Hamilton International Hotel in Dire Dawa, Ethiopia. It is intended for authorized hotel owners, executives, managers, technology partners and implementation teams.

Important

The solution scope, timeline and investment figures in this proposal are planning estimates. Final commitments should be confirmed after on-site discovery, process mapping, infrastructure assessment, tax and payment integration review, and approval of the detailed statement of work.

Document Control

Item	Details
Document title	Enterprise Hotel Management System Proposal
Client	Hamilton International Hotel
Location	Dire Dawa, Ethiopia
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Proposal Purpose

- Define the business vision, system scope and expected operational benefits.
- Present a practical architecture that can start lean and scale safely.
- Provide a phased implementation roadmap with governance, training and acceptance controls.
- Establish a foundation for detailed requirements, pricing and contract negotiation.

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Proposal structure

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EXECUTIVE SUMMARY

A unified digital operating platform for Hamilton International Hotel

Hamilton International Hotel requires more than a basic booking application. The proposed Hotel Management System (HMS) is an enterprise operating platform designed to connect the complete hotel value chain: reservations, front desk, rooms, housekeeping, food and beverage, guest relationships, finance, revenue management, staff administration, integrations and executive reporting.

The proposal recommends a phased architecture. The first release will deliver a stable modular core with strong security, audit trails and clear interfaces. As transaction volume, properties or integrations grow, selected domains can be separated into independent services without replacing the whole platform. This approach reduces early complexity while protecting long-term scalability.

ONE	10+	99.5%	4
Single operational view	Core business modules	Target availability	Implementation phases

Expected Business Outcomes

- Faster reservations, check-in, room allocation, service posting and check-out.
- Real-time room status coordination between front desk and housekeeping.
- Accurate guest folios, payments, invoices, night audit and management reporting.
- Improved guest recognition, preferences, service recovery and repeat-stay engagement.
- Controlled staff access, complete audit logs and reduced revenue leakage.
- A digital foundation for online booking, OTA connectivity, mobile operations and future multi-property expansion.

Recommended Decision

Approve a structured discovery and design phase, followed by a controlled minimum viable release covering reservations, front desk, rooms, housekeeping, guest profiles, billing and management dashboards.

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HOTEL DIGITAL TRANSFORMATION VISION

From disconnected tasks to coordinated hospitality operations

Vision Statement

To equip Hamilton International Hotel with a reliable, secure and intelligent digital platform that gives every authorized team member the information and tools required to deliver excellent guest service, protect revenue and make timely management decisions.

Transformation Principles

Principle	Application to Hamilton International Hotel
Guest-first	Every workflow should reduce waiting, errors and repeated questions for guests.
One source of truth	Reservations, room status, folios, guest data and reports should use controlled shared records.
Mobile-ready operations	Housekeeping, maintenance, supervisors and managers should work effectively on phones and tablets.
Secure by design	Least privilege, audit history, protected secrets, backups and monitoring are included from the start.
Practical scalability	Launch a maintainable core, then scale high-volume or specialized functions independently.
Local operational fit	Support Ethiopian tax, payment, connectivity and staffing realities while remaining internationally extensible.

Target Operating Model

Connected Hotel

The front desk sees live room readiness; housekeeping receives prioritized tasks; restaurant charges post to the correct guest folio; finance reconciles payments; management sees occupancy, revenue, receivables and operational exceptions from one controlled platform.

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BUSINESS OBJECTIVES AND SUCCESS MEASURES

Measurable outcomes for operations, guests and management

Objective	Business Result	Example KPI
Operational control	Digitize core hotel processes and reduce manual handoffs.	Check-in cycle time, room turnaround time, unresolved task count
Revenue protection	Ensure every charge, discount, refund and payment is authorized and traceable.	Unposted charge rate, void/discount exceptions, reconciliation variance
Guest experience	Create faster, more personalized and consistent service.	Guest wait time, complaint resolution time, repeat guest recognition
Management visibility	Provide reliable daily and monthly performance dashboards.	Report preparation time, data completeness, management adoption
Scalability	Prepare for online channels, multiple outlets and future properties.	Integration readiness, performance capacity, configuration reuse
Security and resilience	Protect guest, staff and financial information.	Access review completion, backup success, recovery testing, incident rate

Initial Target Measures

- At least 95% of occupied-room charges posted through controlled digital workflows.
- Housekeeping room-status updates visible to the front desk within one minute under normal connectivity.
- Daily operational and financial summaries available without manual spreadsheet consolidation.
- All privileged user activity retained in tamper-resistant audit logs.
- Successful backup monitoring and documented recovery tests before production acceptance.

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PROPOSED SOLUTION SCOPE

Business capabilities included in the target platform

In-Scope Capabilities

Domain	Scope
Executive & operations	Management dashboard, alerts, occupancy, revenue, arrivals, departures and exceptions
Reservations	Availability, rates, bookings, groups, companies, deposits, amendments and cancellations
Front desk	Pre-arrival, registration, room assignment, check-in, stay management, room move and check-out
Rooms & housekeeping	Room inventory, status, cleaning tasks, inspections, out-of-order, maintenance requests and minibar
Guest CRM	Profiles, preferences, history, companions, documents, consent, feedback and service recovery
Food & beverage	Outlets, menu, POS orders, kitchen status, payments, room posting and end-of-shift controls
Finance	Guest folios, invoices, receipts, cashiers, refunds, taxes, receivables, night audit and exports
Revenue	Rate plans, restrictions, forecasting, pickup, segmentation and pricing recommendations
Administration	Users, roles, shifts, departments, configuration, approvals, audit and system health
Notifications & integration	Email, SMS, push, payment gateway, OTA/channel manager, accounting and identity integrations

Scope Boundaries for Phase 1

- One hotel property with configurable room types, rooms, outlets and departments.
- Web application optimized for desktop and mobile browsers; selected mobile workflows can be packaged later as native apps.
- Core integrations prioritized by business value and technical readiness.
- Historical data migration limited to agreed master data, open reservations, in-house guests, balances and selected guest history.

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FUNCTIONAL MODULES

Detailed module capabilities

Module	Key Functions
Dashboard	KPIs, arrivals/departures, room status, occupancy, revenue, collections, alerts and drill-downs.
Reservations	Calendar, availability search, rate quote, guarantees, deposits, groups, waitlist, cancellation and no-show.
Front Desk	Registration cards, identity capture, room assignment, key reference, wake-up, messages, room moves and checkout.
Rooms & Housekeeping	Clean/dirty/inspected status, assignments, priority queues, linen/minibar notes, defects and supervisor inspection.
Guest CRM	Single guest profile, stay history, preferences, VIP/blacklist flags, loyalty reference, consent and feedback.
Food & Beverage	Table/service orders, room charge, cash/card/mobile payment, void controls, kitchen tickets and outlet reports.
Finance	Folios, split billing, company billing, invoices, receipts, tax, cash shifts, deposits, refunds and night audit.
Revenue	Rate plans, seasons, occupancy rules, restrictions, pickup, forecast and recommendation workflow.
Admin	Roles, user lifecycle, approvals, configuration, audit, backups, integration keys and operational health.
Notifications	Booking confirmation, pre-arrival, payment, stay, checkout, feedback and internal exception alerts.

Configuration over Custom Code

Room types, taxes, rate plans, policies, departments, outlets, document templates, approval limits and notification templates should be configurable by authorized administrators wherever practical.

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USER ROLES AND ACCESS MODEL

Role-based access with segregation of duties

Role	Typical Access
Owner / Executive	Read-only enterprise view, approvals, strategic reports and audit oversight
General Manager	Full operational visibility, selected approvals, configuration oversight and reports
Front Office Manager	Reservations, desk operations, room allocation, cashier supervision and exceptions
Receptionist / Cashier	Assigned reservation, check-in/out, folio and payment functions within shift limits
Housekeeping Manager	Assignments, inspections, room status, productivity and supply reporting
Housekeeper	Assigned tasks, status updates, notes, minibar and issue reporting
F&B Manager	Outlet setup, menus, prices, void approval, stock linkage and reporting
Waiter / Cashier	Orders, settlements and room postings within assigned outlet and shift
Finance / Auditor	Invoices, reconciliation, receivables, night audit, exports and financial reports
Revenue Manager	Rates, restrictions, forecasts, pickup analysis and approved recommendations
System Administrator	Identity, roles, configuration, integrations, monitoring and technical support

Access Control Rules

- Least privilege by role, department, outlet, property and shift.
- Maker-checker approval for sensitive refunds, discounts, rate overrides and master-data changes.
- Multi-factor authentication for privileged and remote administrative accounts.
- Automatic session timeout, password controls, failed-login protection and user deactivation workflow.
- Audit evidence records who performed an action, when, from where and what changed.

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GUEST JOURNEY AND OPERATIONAL WORKFLOWS

End-to-end service coordination

Core Guest Journey

Stage	Digital Workflow
1. Discover & Book	Direct web, phone, walk-in, corporate or OTA reservation; availability and rate validation.
2. Pre-Arrival	Confirmation, deposit status, preferences, transport notes, room planning and arrival list.
3. Arrival & Check-In	Identity verification, registration, room assignment, payment guarantee and key reference.
4. In-House Stay	Requests, messages, housekeeping, maintenance, F&B room charges, extensions and room moves.
5. Settlement & Check-Out	Folio review, split billing, payment, invoice/receipt, room status transition and feedback request.
6. Post-Stay Relationship	Guest history, preferences, feedback, service recovery and targeted return communication.

Critical Cross-Department Workflows

- Check-out changes room to dirty and automatically creates a housekeeping task.
- Housekeeping completion sends room to inspection or clean status based on hotel policy.
- Maintenance defect can place a room out-of-order and block it from availability.
- Restaurant room charge validates guest, room, folio and posting authority before confirmation.
- Night audit closes operational date, validates open folios and produces exception reports.
- Manager alerts surface overdue rooms, unpaid deposits, high discounts, failed integrations and reconciliation issues.

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SOLUTION ARCHITECTURE

Scalable, secure and maintainable platform design

Recommended Architecture Pattern

The recommended starting point is a modular service-oriented core behind an API gateway. Each business domain has clear interfaces, data ownership and deployment boundaries. High-growth services can later be extracted into independent microservices when transaction volume, team size or integration complexity justifies the operational cost.

Guest Web / Booking	Staff Web Application	Mobile Operations
API Gateway	Identity & Access	Notification Gateway
Reservations	Front Desk / Stay	Rooms / Housekeeping
Guest CRM	Food & Beverage	Finance / Billing
Revenue & Analytics	Integrations	Administration / Audit
PostgreSQL Transaction Data	Cache / Queue	Object & Backup Storage
Cloud / Data Center Infrastructure	Monitoring & Logs	Disaster Recovery

Figure 1. Logical architecture layers for the proposed HMS.

Key Technical Principles

- API-first interfaces using versioned REST endpoints and documented events.
- Transactional consistency for reservations, inventory, folios and payments.
- Asynchronous events for notifications, analytics updates and external integrations.
- Central identity with service-level authorization and zero-trust network rules.
- Observability through structured logs, metrics, traces, alerts and business audit records.

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APPLICATION AND REPOSITORY STRUCTURE

Organized codebase for delivery and long-term maintenance

The proposed repository structure is suitable for an enterprise program, but it should be implemented in stages. Empty microservices should not be created only for appearance; each service should have a clear business owner, data boundary, operational need and support plan.

Recommended Top-Level Structure

Path	Purpose
docs/	API specifications, architecture decisions, database models, deployment runbooks and security controls
frontend/web/	Primary responsive staff and management web application
frontend/mobile/	Optional native or cross-platform application for mobile workflows
backend/	API gateway, domain services and shared libraries
database/	Version-controlled migrations and controlled seed data
infrastructure/	Container, Kubernetes, Terraform and monitoring configuration
tests/	Unit, integration, end-to-end, security and performance tests
scripts/	Repeatable build, migration, backup and deployment automation
.github/workflows/	Continuous integration, security scanning and controlled deployment pipelines

Recommended Technology Baseline

Layer	Baseline
Web frontend	React + TypeScript, responsive component library and controlled state management
Backend	Java Spring Boot or a consistent enterprise backend selected during discovery
Database	PostgreSQL with versioned migrations and audited backup procedures
API documentation	OpenAPI / Swagger
Deployment	Docker; managed cloud or hybrid infrastructure; Kubernetes only when operationally justified
Testing	JUnit, API integration tests and Playwright/Cypress end-to-end tests

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DATA ARCHITECTURE

Trusted hotel records, controlled ownership and auditability

Data Domain	Primary Entities
Property & configuration	Property, building, floor, room type, room, outlet, department, tax, currency and policy
Reservations	Reservation, reservation room, rate plan, inventory, guarantee, deposit, cancellation and group block
Guests	Guest, address, identification, preference, consent, companion, company and stay history
Stay operations	Registration, room assignment, key reference, service request, message, room move and wake-up
Housekeeping	Room status, task, assignment, inspection, defect, linen/minibar record and out-of-order period
F&B	Outlet, menu item, order, order item, kitchen status, settlement and room-charge posting
Finance	Folio, folio line, invoice, receipt, payment, refund, cashier shift, ledger mapping and audit batch
Administration	User, role, permission, approval, audit event, integration credential reference and system setting

Data Governance Rules

- Unique identifiers for guests, reservations, stays, folios, invoices and payments.
- No hard deletion of financial and audit evidence; use controlled reversals and retention policies.
- Optimistic locking or equivalent controls to prevent conflicting room and folio updates.
- Encryption for data in transit and at rest; especially identity, contact and payment-related information.
- Defined retention, archival and anonymization rules for guest and operational data.
- Read replicas or analytics pipelines should not become the source of operational truth.

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INTEGRATION ARCHITECTURE

Controlled connection to hospitality and local services

Integration	Information Exchange
Online Travel Agencies / Channel Manager	Rates, availability, reservations, modifications and cancellations
Payment Services	Card, mobile money, bank transfer reference, payment confirmation and refund status
SMS / Email / Push	Guest confirmation, reminders, receipts, internal alerts and service communication
Accounting / ERP	Chart-of-account mapping, daily revenue, taxes, receivables, payments and journals
Door Lock / Key System	Key issuance reference and room authorization where supported
POS / Kitchen / Inventory	Outlet sales, kitchen tickets, stock consumption and room posting
Identity / Document Devices	Document scan or registration device where legally and operationally appropriate
Business Intelligence	Controlled export or pipeline for management analytics

Integration Control Standards

- Use documented APIs, signed webhooks, idempotency keys and retry policies.
- Store credentials in a secrets manager, never in source code or public configuration files.
- Queue failed messages and provide an operations dashboard for retries and reconciliation.
- Separate external reference numbers from internal identifiers.
- Record request/response metadata safely without exposing sensitive payment or identity data in logs.

Connectivity Resilience

Critical front-desk and cashier functions should use graceful retry and clear offline/failed-state handling. External connectivity failure must not silently create duplicate reservations or duplicate charges.

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SECURITY, PRIVACY AND COMPLIANCE

Protection of guests, revenue and hotel operations

Control Area	Proposed Controls
Identity	Strong passwords, MFA for privileged users, account lockout, lifecycle management and SSO readiness
Authorization	Role and permission controls, property/outlet scope, approval limits and segregation of duties
Application security	Input validation, secure sessions, CSRF protection, rate limiting and secure API design
Data protection	TLS, encrypted storage, secrets management, masked sensitive fields and controlled exports
Financial controls	No storage of full card details, authorized refunds/voids, cashier shifts and reconciliation evidence
Audit	Immutable business audit events for logins, configuration, rates, room moves, discounts, refunds and payments
Infrastructure	Network segmentation, patched images, vulnerability scanning, protected backups and endpoint hardening
Monitoring	Security alerts, anomaly review, failed-login trends, privileged activity and incident procedures

Privacy Principles

- Collect only data needed for reservation, stay, legal, service and legitimate relationship purposes.
- Limit access to guest identity and contact information based on job responsibility.
- Provide retention, correction and controlled disclosure procedures aligned with applicable requirements.
- Separate consent-based marketing preferences from operational communication requirements.
- Document incident response, notification and evidence-preservation responsibilities.

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INFRASTRUCTURE AND DEPLOYMENT

Reliable operation with controlled environments and recovery

Environment Model

Environment	Purpose
Development	Developer testing with synthetic data and no production credentials
Test / QA	Integration, end-to-end and regression testing with production-like configuration
Staging	Release candidate validation, migration rehearsal and user acceptance testing
Production	Hardened live environment with monitoring, backups, restricted access and change approval

Deployment Options

Option	Consideration
Managed cloud	Fast provisioning, managed database, scalable storage, monitoring and disaster-recovery options.
Hybrid	Core cloud services with selected local integrations or contingency capabilities at the hotel.
On-premises	Greater local control but higher responsibility for power, cooling, hardware, security, backup and recovery.

Recommended Operational Controls

- Containerized services with repeatable infrastructure configuration and automated deployment checks.
- Daily encrypted backups, point-in-time recovery where supported and regular restore testing.
- Monitoring for application availability, API latency, database health, disk use, queue failures and integration errors.
- Documented recovery time and recovery point objectives confirmed during discovery.
- Release rollback, emergency change and incident escalation procedures.

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REPORTING, ANALYTICS AND AI READINESS

From operational visibility to intelligent decision support

Report Group	Examples
Daily Operations	Occupancy, arrivals, departures, stayovers, room status, out-of-order and unresolved tasks
Revenue	Room revenue, ADR, RevPAR, occupancy, pickup, segment, channel, package and discount analysis
Finance	Payments, cashier shifts, tax, invoices, receivables, refunds, voids, deposits and reconciliation
Guest	Repeat guests, VIPs, preferences, feedback, complaint trends, source market and company production
Housekeeping	Room turnaround, productivity, inspection failures, maintenance defects and task ageing
F&B	Outlet sales, average check, payment mix, voids, discounts, room postings and item performance
Audit & Security	Access changes, failed logins, privilege use, rate override, refund, export and configuration events

AI-Ready Capabilities - Later Phase

- Occupancy and revenue forecasting using clean historical and future reservation data.
- Rate recommendations with transparent rules, confidence and manager approval.
- Guest preference suggestions based on consented history and service records.
- Anomaly alerts for unusual discounts, refunds, voids or reconciliation variances.
- Natural-language management summaries generated from controlled metrics, not raw unrestricted data.

Governance First

AI recommendations should be explainable, reviewed by authorized managers and monitored for accuracy. The operational system of record remains authoritative.

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IMPLEMENTATION METHODOLOGY

Discover, design, build, validate and stabilize

Stage	Main Activities
1. Discovery	Stakeholder interviews, process mapping, current-system review, data assessment, integration inventory and risk baseline
2. Solution Design	Requirements, workflow prototypes, architecture, data model, security design, backlog and acceptance plan
3. Iterative Build	Prioritized configuration and development in short cycles with demonstrations and feedback
4. Validation	Integration, security, performance, migration rehearsal and user acceptance testing
5. Go-Live	Cutover, production verification, support desk, issue triage and management reporting
6. Stabilization	Hypercare, performance tuning, adoption review, backlog closure and handover

Delivery Principles

- Business owners approve workflows and acceptance criteria before development is considered complete.
- Working software is demonstrated regularly; risks and decisions are documented.
- Security, data migration, training and operations are continuous workstreams, not end-stage tasks.
- Changes are prioritized through a governed backlog and assessed for schedule and cost impact.
- Production readiness requires evidence: test results, backups, monitoring, runbooks, training and approved cutover.

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DELIVERY ROADMAP AND MILESTONES

Illustrative phased plan subject to discovery

Phase	Indicative Duration	Main Deliverables
Phase 0 - Discovery & Blueprint	3-4 weeks	Approved requirements, process maps, architecture, backlog, migration and rollout plan
Phase 1 - Core PMS	10-14 weeks	Reservations, front desk, rooms, housekeeping, guest profiles, folios, payments and dashboards
Phase 2 - Operations Expansion	6-10 weeks	F&B, notifications, company billing, improved reports, mobile housekeeping and priority integrations
Phase 3 - Revenue & Integration	6-10 weeks	Revenue tools, OTA/channel, accounting export, advanced analytics and automation
Phase 4 - Optimization	Ongoing	Performance, AI-assisted insights, loyalty, self-service and multi-property readiness

Key Milestones

1. Project charter, governance and discovery kickoff approved.
2. Target workflows, data model and security architecture signed off.
3. Core PMS demonstration completed with agreed end-to-end scenarios.
4. Migration rehearsal and user acceptance exit criteria achieved.
5. Production readiness review passed and go-live authorized.
6. Stabilization exit and operational handover accepted.

Schedule Dependency

The final timeline depends on stakeholder availability, integration providers, data quality, infrastructure readiness, scope stability and speed of acceptance decisions.

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TESTING AND QUALITY ASSURANCE

Evidence-based readiness before production

Test Level	Coverage
Unit testing	Business rules, calculations, validation and service logic
API & integration testing	Contracts, authentication, retries, duplicate prevention and error handling
End-to-end testing	Reservation through check-out, room turnaround, F&B posting, night audit and reporting
Security testing	Authorization, session controls, vulnerabilities, secrets, logging and privilege abuse scenarios
Performance testing	Availability search, check-in, posting, dashboards, night audit and concurrent-user loads
Migration testing	Record counts, balances, open reservations, guest profiles and reconciliation
User acceptance testing	Department-approved scenarios using realistic roles and operating procedures
Recovery testing	Backup restore, service recovery, rollback and integration-failure procedures

Quality Gates

- No unresolved critical or high-severity defect at production launch unless formally accepted with mitigation.
- Financial and inventory workflows reconcile to agreed test cases.
- Role-based access tests confirm users cannot perform unauthorized operations.
- Performance meets agreed response targets under expected load.
- Runbooks, monitoring, backup and rollback evidence are approved.

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DATA MIGRATION AND CUTOVER

Controlled transition with reconciliation and fallback

Migration Scope Candidates

- Room types, rooms, rate plans, taxes, packages, companies, travel agents, users and configuration.
- Future reservations, deposits, group blocks and relevant notes.
- Current in-house guests, room assignments, folio balances and payment references.
- Selected guest profiles, preferences and stay history where quality and consent permit.
- Open company receivables and approved finance opening balances.

Migration Process

Step	Control
Extract	Obtain source exports, file definitions and record counts.
Cleanse	Remove duplicates, normalize formats, resolve invalid codes and identify missing mandatory data.
Transform	Map source fields to approved HMS entities and reference values.
Load	Use repeatable migration scripts with logging and error reports.
Reconcile	Compare counts, totals, balances and sampled records with business owners.
Approve	Formal sign-off before final production cutover.

Cutover Controls

- Freeze window and clear responsibility matrix.
- Final backup and recovery checkpoint before migration.
- Department-by-department verification checklist.
- Rollback criteria and decision authority.
- Enhanced support coverage during first operational cycles and night audit.

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TRAINING, SUPPORT AND CHANGE MANAGEMENT

Adoption as a core project deliverable

Audience	Training Focus
Executive & managers	Dashboards, approvals, exceptions, reports, governance and decision routines
Reservations & front desk	Availability, booking, check-in/out, room moves, folios, payments and service recovery
Housekeeping	Assignments, room status, defects, inspections and mobile workflow
F&B	Orders, settlements, room posting, void/discount rules and shift close
Finance & audit	Invoices, reconciliation, receivables, cashier controls, night audit and exports
System administrators	Users, roles, configuration, monitoring, integrations, backup and support process

Support Model

- Tier 1: hotel super-users handle common questions and basic operating issues.
- Tier 2: application support handles configuration, data correction and functional incidents.
- Tier 3: engineering handles defects, performance, integrations and security incidents.
- Service desk records severity, ownership, response, resolution and root-cause actions.

Change Management Activities

- Department champions and early prototype demonstrations.
- Updated standard operating procedures aligned with the digital workflow.
- Role-based training, practice environment and competency checks.
- Go-live communication, floor support and adoption dashboards.

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GOVERNANCE AND PROJECT TEAM

Clear accountability, decisions and escalation

Role	Responsibility
Executive Sponsor - Hotel	Owns business case, resolves major issues and approves strategic scope
Hotel Product Owner	Prioritizes requirements, coordinates departments and accepts business outcomes
Project Manager	Controls plan, dependencies, budget, risks, decisions and reporting
Department Leads	Define workflows, validate requirements, test and lead adoption
Solution Architect	Owns architecture, integration, security and technical quality
Engineering Team	Builds, configures, tests and documents the platform
Data & Migration Lead	Owns data mapping, cleansing, load and reconciliation
Security / Infrastructure Lead	Owns environments, monitoring, access, backup and incident readiness
Training & Support Lead	Owns training, support readiness, knowledge and hypercare

Governance Rhythm

- Weekly delivery review: progress, demos, blockers, defects and next priorities.
- Biweekly steering review: scope, schedule, budget, risks and decisions.
- Architecture and security reviews at major design and release gates.
- Formal change control for requirements affecting cost, timeline, integrations or data migration.
- Decision log and risk register maintained throughout the program.

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RISKS AND MITIGATION

Proactive management of delivery and operational risk

Risk	Impact	Mitigation
Unclear or changing requirements	High	Structured discovery, prototypes, signed acceptance criteria and change control
Poor source data quality	High	Early profiling, cleansing owners, migration rehearsals and reconciliation
Integration provider delays	Medium/High	Prioritize early, use mock interfaces, define fallback/manual procedures
Internet or power instability	Medium/High	Resilient hosting, local network readiness, retries, UPS and contingency runbooks
Low user adoption	High	Department champions, SOP updates, role-based training and floor support
Excessive early microservices	Medium	Start with modular core; separate services only for justified scale or ownership
Security misconfiguration	High	Least privilege, secrets manager, reviews, scanning, monitoring and regular access review
Revenue reconciliation issues	High	Controlled posting rules, cashier shifts, nightly controls, parallel validation and audit
Vendor dependency	Medium	Documentation, source control, open standards, knowledge transfer and exit provisions

Risk Ownership

Each material risk should have an owner, response action, due date, status and escalation threshold. Risks should be reviewed at every steering meeting and at production-readiness gates.

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COMMERCIAL PLANNING AND INVESTMENT MODEL

Transparent estimation before final quotation

A final fixed price cannot be responsibly confirmed from a repository outline alone. Commercial commitment should follow discovery, approved scope, integration assessment, migration profiling and infrastructure decisions. The model below provides a structured basis for budgeting.

Cost Component	What Drives the Cost
Discovery & solution design	Workshops, process mapping, requirements, architecture, prototypes and statement of work
Core platform implementation	Reservations, front desk, rooms, housekeeping, guest, finance and dashboard
Operations modules	F&B, notifications, revenue, advanced reporting and mobile workflows
Integrations	Payment, OTA/channel, accounting, SMS/email, door lock or other providers
Data migration	Extraction support, cleansing, transformation, load, reconciliation and cutover
Infrastructure	Hosting, database, storage, monitoring, backups, domains, certificates and network readiness
Training & go-live	Materials, sessions, practice, onsite/remote support and stabilization
Ongoing support	Service desk, maintenance, monitoring, backups, updates and improvement capacity

Indicative Budget Bands for Planning Only

Planning Item	Illustrative Range
Discovery & blueprint	ETB 450,000 - 900,000
Core PMS implementation	ETB 3,500,000 - 7,500,000
Integrations and advanced modules	ETB 1,500,000 - 5,000,000+
Training, migration and go-live	ETB 700,000 - 2,000,000
Annual hosting and support	Typically 15% - 25% of implementation value, plus third-party fees

Commercial Note

These ranges are non-binding planning assumptions and exclude taxes, third-party licenses, payment/OTA transaction fees, hardware, networking and major custom device integrations unless explicitly included in the final statement of work.

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ACCEPTANCE CRITERIA AND NEXT STEPS

From proposal review to approved implementation

High-Level Acceptance Criteria

Category	Acceptance Requirement
Functional	Approved end-to-end scenarios complete successfully for every in-scope department
Financial	Folio, tax, payment, invoice, refund and night-audit test cases reconcile
Rooms	Availability, room status, assignment, room move and out-of-order controls behave correctly
Security	Role access, approvals, privileged actions, audit and session controls pass agreed tests
Performance	Critical workflows meet approved response and concurrency targets
Migration	Counts, balances, open stays/reservations and sampled records are reconciled and signed off
Operations	Monitoring, backup, restore, runbooks, support and escalation are ready
Adoption	Key user groups complete training and designated super-users demonstrate competency

Recommended Immediate Next Steps

7. Nominate the executive sponsor, hotel product owner and department representatives.
8. Approve a 3-4 week discovery and solution-blueprint engagement.
9. Inventory current reservations, room master, guest data, finance processes and existing systems.
10. Confirm priority integrations, infrastructure preference and operating constraints in Dire Dawa.
11. Produce the final requirements specification, phased backlog, architecture and fixed commercial proposal.
12. Approve the statement of work and begin Phase 1 delivery.

Management Approval Requested

Approve progression to the discovery and solution-blueprint phase for Hamilton International Hotel, Dire Dawa, Ethiopia.

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APPENDIX - PROPOSED REPOSITORY STRUCTURE

Improved enterprise project organization

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hotel-management-system/
|-- docs/
|   |-- api/           # OpenAPI specifications and examples
|   |-- architecture/  # Architecture decisions and diagrams
|   |-- database/      # ER models, ownership and data dictionary
|   |-- deployment/    # Runbooks, rollback, backup and recovery
|   |-- security/      # Threat model, controls and compliance evidence
|-- frontend/
|   |-- web/
|       |-- src/
|           |-- app/      # Routing, providers and application shell
|           |-- components/ # Shared design-system components
|           |-- features/  # Domain modules by business capability
|           |-- services/  # Typed API clients
|           |-- store/     # State management
|           |-- hooks/     # Reusable hooks
|           |-- utils/     # Shared utilities
|       |-- public/
|       |-- package.json
|   |-- mobile/         # Optional React Native / Flutter app
|-- backend/
|   |-- api-gateway/
|   |-- auth-service/
|   |-- reservation-service/
|   |-- room-service/
|   |-- guest-service/
|   |-- f-b-service/
|   |-- finance-service/
|   |-- revenue-service/
|   |-- notification-service/
|   |-- integration-service/
|   |-- common/         # Small, governed shared libraries only
|-- database/
|   |-- migrations/
|   |-- seeds/
|-- infrastructure/
|   |-- containers/
|   |-- kubernetes/     # Adopt when operationally justified
|   |-- terraform/
|   |-- monitoring/
|-- tests/
|   |-- unit/
|   |-- integration/
|   |-- e2e/
|   |-- security/
|   |-- performance/
|-- scripts/
|-- .github/workflows/
|-- docker-compose.yml
|-- README.md
|-- LICENSE

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APPENDIX - INITIAL DATA ENTITIES AND APIS

Starting point for detailed technical design

Initial API Groups

API Group	Purpose
/api/v1/auth	Login, MFA, session, password, user and role administration
/api/v1/availability	Room availability, restrictions, rate plans and quote calculation
/api/v1/reservations	Create, amend, guarantee, cancel, group and no-show workflows
/api/v1/frontdesk	Arrivals, registration, room assignment, check-in, stay actions and check-out
/api/v1/rooms	Room master, status, housekeeping tasks, inspections, defects and out-of-order
/api/v1/guests	Profiles, identity references, preferences, history, consent and feedback
/api/v1/folios	Folios, charges, adjustments, invoices, receipts, payments and refunds
/api/v1/fb	Outlets, menu, orders, kitchen status, settlements and room postings
/api/v1/revenue	Rates, restrictions, forecasts, pickup and recommendations
/api/v1/reports	Operational, financial, guest, housekeeping, F&B and audit reports
/api/v1/integrations	Webhooks, channel, payment, messaging and accounting connectors

Important Cross-Cutting Fields

- property_id, department_id and outlet_id for future organizational scope.
- created_at, updated_at, created_by, updated_by and version for traceability and concurrency.
- status with controlled transition rules rather than unrestricted free text.
- external_reference and idempotency_key for reliable integrations.
- currency, tax_code and business_date for correct hotel and financial processing.
- audit_context for approval, reason, source device and correlation reference where applicable.

HAMILTON INTERNATIONAL HOTEL

DIRE DAWA, ETHIOPIA

ENTERPRISE HOTEL MANAGEMENT SYSTEM

A practical digital foundation for guest service, operational control and sustainable growth.

Proposal Status

Prepared for management review. The recommended next action is approval of the discovery and solution-blueprint phase.

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